

Modularity Maximization: An Experimental Survey

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Abstract

Community detection methods commonly optimize modularity, an objective function designed to maximize intra-community connectivity while penalizing inter-community connectivity relative to random expectation. Over the past two decades, a wide range of algorithms have been proposed to optimize modularity, spanning classical heuristic and exact methods as well as more recent machine learning-based approaches that incorporate node attributes. Yet, survey literature and experimental evaluations remain fragmented, often confined to narrow subsets of algorithms or a single paradigm, limiting insight into relative strengths and trade-offs in practice. In this work, we present the first systematic experimental survey of modularity-based graph clustering algorithms that jointly evaluates classical and learning-based approaches. We study 17 representative algorithms spanning greedy, multilevel, metaheuristic, streaming, exact, and neural paradigms, and organize them within a unified taxonomy grounded in their algorithmic design principles. Through a consistent experimental evaluation across multiple datasets and evaluation criteria, we analyze empirical trade-offs in solution quality and scalability, providing comparative insight rather than identifying a single best method. We additionally contribute a unified, open-source framework for the implementation of all 17 evaluated clustering algorithms, supporting reproducibility and future benchmarking.

CCS Concepts

• **Theory of computation** → **Graph algorithms analysis**; • **Mathematics of computing** → **Graph theory**; • **Computing methodologies** → **Cluster analysis**.

Keywords

Community detection, Graph clustering, Modularity optimization, Experimental evaluation

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1 Introduction

Networks are useful in modeling complex systems across the natural sciences, social sciences, and engineering. Such systems are naturally represented as graphs, where vertices denote entities and edges encode interactions between them. Prominent examples include online social networks, communication infrastructures, biological interaction networks, financial systems, and information networks. A common structural property of many real-world graphs is the presence of *community structure*, where vertices organize into groups that are more densely connected internally than externally.

The task of identifying these groups, commonly referred to as *graph clustering* or *community detection*, is central to network analysis. Community detection supports a wide range of applications, including the analysis of social and political behavior, fraud detection [1], protein interaction analysis and gene expression [8, 28], program optimization [11, 16], and epidemic modeling [18]. Beyond application-specific insights, community structure offers a lens into the organization of networks, highlighting cohesive substructures, boundary vertices, and inter-community connectivity patterns. Despite extensive research, it is well understood that no single clustering method dominates across all network types, motivating continued investigation into algorithmic design and comparative evaluation.

Most community-detection algorithms model clustering as an optimization problem over partitions of the vertex set, assigning a quality score to each candidate clustering. The goal is to identify a partition that optimizes this score, often under stringent computational constraints. For many commonly used objectives, finding an optimal clustering is NP-hard, and practical methods therefore rely on heuristic or approximate algorithms.

Among the many objectives proposed for graph clustering, *modularity*, introduced by Girvan and Newman [13], remains one of the most widely adopted. Modularity quantifies the density of edges within clusters compared to those between clusters, while accounting for random chance, with the goal of maximizing densely connected communities while penalizing excessive inter-community connectivity [13, 19]. Optimizing modularity is strongly NP-complete, and consequently a wide variety of heuristic algorithms have been proposed. Given the common use of modularity in the literature, this survey focuses primarily on modularity-based approaches, though we include a small number of widely used non-modularity community-detection baselines. While alternative paradigms, such as spectral clustering or probabilistic generative models, are important in the broader literature, modularity provides a common

objective that enables meaningful comparison across different algorithmic strategies.

Within modularity-based community detection, two broad classes of algorithms are identified. *Classical* approaches rely on discrete optimization heuristics that operate on graph topology. Their clustering decisions are driven solely by graph structure and a well-defined objective function, yielding transparent optimization criteria and interpretable outcomes. These methods include greedy merging, local vertex-move strategies, multilevel coarsening, metaheuristics, streaming algorithms, and exact formulations. *Learning-based* approaches integrate modularity into deep learning pipelines, most commonly through graph neural networks. These methods are motivated by the ability to incorporate node attributes, embeddings, or other side information alongside graph topology, through feature representations. This increased expressiveness introduces different trade-offs in terms of interpretability, training complexity, and computational cost.

Several challenges continue to shape the design and evaluation of clustering algorithms, driven by the scale, complexity and frequently-updating content of real-world graphs. Clustering algorithms must balance high-solution quality in contexts where data may be noisy or ground-truth communities unknown, with scalability in memory usage and runtime. Additionally, many algorithms are sensitive to pre-set parameters, such as assumed cluster counts, which complicates deployment in practice. There remains a need to systemally address how existing clustering algorithms balance trade-offs given these challenges.

Although numerous community-detection algorithms and surveys exist, there is currently no comprehensive experimental study that evaluates classical and learning-based modularity optimization methods together under a unified framework. There are limited claims about their relative strengths in the literature, and those that exist vary, particularly regarding scalability. This paper addresses this gap by providing a systematic experimental survey of modularity-based graph clustering algorithms across both classical and learning-based paradigms. Rather than advocating a single best method, this survey aims to clarify the landscape of modularity-based clustering by highlighting the strengths, limitations, and trade-offs of different algorithmic approaches across diverse settings.

In this study, we contribute the following:

- A taxonomy of modularity-based clustering algorithms, organizing methods by algorithmic paradigm and optimization strategy.
- A systematic review of 18 representative algorithms, covering greedy, multilevel, metaheuristic, streaming, exact, and learning-based approaches.
- The first large-scale experimental comparison of classical and learning-based modularity optimization methods, analyzing their relative performance, scalability, and trade-offs.
- A unified open-source experimental framework, including the adaptation of all evaluated algorithms behind a common interface, to facilitate future benchmarking and reproducibility in graph clustering research.

2 Preliminary

We use the term algorithm to denote a concrete computational procedure. Except where explicitly stated, the algorithms considered here are heuristics for modularity maximization and do not guarantee optimality.

Let $G = (V, E)$ be an undirected simple graph, with vertex set V and edge set E . We use $n := |V|$ and $m := |E|$ to denote the number of vertices and edges. The *neighborhood* of u is $N(u) := \{v : \{u, v\} \in E\}$ and the *degree* is $d(u) := |N(u)|$. A *clustering* C of a graph is the partition of the nodes V into k disjoint blocks $V_1, \dots, V_k$ such that the union $V_1 \cup \dots \cup V_k = V$.

The *modularity* score of a clustering is the fraction of edges that are within one cluster minus the expected fraction if the edges were distributed at random. Let $L_i := |\{u, v\} \in E : \{u, v\} \subseteq V_i|$ be the set of edges within block i . With that, we can define modularity as:

$$Q(C) := \sum_{i=1}^k \left(\frac{L_i}{m} - \left(\sum_{u \in V_i} \frac{d(u)}{2m} \right)^2 \right)$$

3 Related Work

The overview and taxonomy of the clustering algorithms included in this survey follow in Section 4. First, we provide a brief overview of relevant objective functions, algorithms, and survey papers. The main motivation for this survey, and reason for focusing primarily on modularity optimization, is highlighted in Figure 1. Despite high interest from multiple scientific communities and the rapid development of new methods, comparisons between methods remain fragmented. This is a problem for both practitioners who want the best algorithm and for future researchers who need to decide which baseline algorithms to use.

Mention WCD, ME, Conductance, Performance, Coverage, NMI, F1, etc. Other surveys. DIMACS Implementation challenge. Related methods that did not make it in. Editing distance.

4 Methods

This section briefly summarizes the clustering algorithms evaluated in our experimental survey. We organize methods primarily by *optimization paradigm*, distinguishing between *classical* combinatorial optimization methods and *learning-based* approaches with trainable parameters. Classical methods are further grouped according to their dominant optimization mechanism (e.g., greedy agglomeration, multilevel local optimization, and metaheuristics/ensembles), as well as by computational setting or guarantees (streaming algorithms and exact solvers). Learning-based methods are grouped by the role modularity plays in training, distinguishing between methods that directly optimize differentiable relaxations of modularity and those that use modularity as a guiding signal within contrastive or infomax objectives. Within each subgroup, algorithms are listed chronologically. Unless stated otherwise, methods aim to maximize modularity; we additionally include a small number of widely used non-modularity baselines (e.g., random-walk- and flow-based methods) for contextual comparison.

4.1 Classical Modularity Optimization Methods

4.1.1 Greedy agglomeration and multilevel local search.

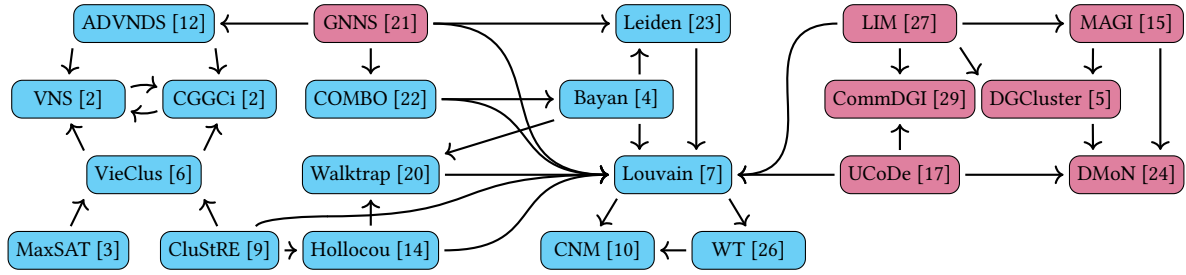


Figure 1: Previous experimental comparisons visualized. Each directed edge indicates that the source algorithm used the destination algorithm in the experimental evaluation. The colors indicate either a learning-based method when red, or a classical optimization algorithm when blue.

Table 1: Comparison of the methods included in the experimental survey. The *Code Available* field is defined as follows: $\times$ means not available, one $\star$ means code is available but not prepared for use in any way, $\star\star$ is for methods with open source implementations with well defined input/output formats, and $\star\star\star$ means the algorithm is easy to use via portable Python packages such as the CDlib package.

Method	Year	Code Available	Stochastic	Parallel	Learning-Based	Features	Fixed $ C $	Exact	Approx.	Running Time
CNM	2004	$\star\star\star$								$O(n \log^2 n)$
Walktrap	2005	$\star\star\star$	$\checkmark$							$O(n^2 \log n)$
WT	2007	$\star\star\star$								Improved CNM
Louvain	2008	$\star\star\star$	$\checkmark$							?
CGGCI	2012	$\times$	?							?
VNS	2012	$\times$	$\checkmark$							?
COMBO	2014	$\star\star\star$	$\checkmark$							?
ADVNDs	2017	$\times$	$\checkmark$							Search-based
HollocoU	2017	$\star\star$	?							?
VieClus	2018	$\star\star$	$\checkmark$	$\checkmark$						?
Leiden	2019	$\star\star\star$	$\checkmark$	$\checkmark$						?
DMoN	2020	$\star$	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$			?
CommDGI	2020	$\star$	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$			?
GNNs	2022	$\star$	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$				?
UCoDe	2023	$\star$	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$			?
DGCluster	2023	$\star$	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$				?
MAGI	2024	$\star$	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$			?
Bayan	2024	$\star\star$						$\checkmark$		?
CluStRE	2025	$\star\star$	?							?
LIM	2025	$\star$	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$				?
MaxSAT	2025	$\times$						$\checkmark$		?

CNM (Clauset–Newman–Moore). CNM is a hierarchical agglomerative algorithm that greedily maximizes modularity, starting from singleton communities (each vertex assigned to its own cluster) and repeatedly merging pairs of communities with the largest modularity gain ΔQ . To achieve scalability, CNM computes $\Delta Q_{c_i, c_j}$ only for community pairs (c_i, c_j) connected by at least one edge, and stores these values in balanced binary trees and max-heaps, allowing efficient updates after each merge. The algorithm produces a dendrogram that represents a hierarchical decomposition of the network into communities at all levels, and returns the clustering attaining maximum modularity. Its running time is $O(md \log n)$, where d is the dendrogram depth.

WT (Wakita–Tsurumi). WT is a refinement of CNM, motivated by the observation that CNM can produce highly unbalanced merges that degrade scalability on large-scale graphs. The algorithm preserves the hierarchical agglomerative structure of CNM, but modifies the merge-selection rule to consider both the modularity gain $\Delta Q_{c_i, c_j}$ and a community-size ratio $\text{ratio}(c_i, c_j)$. This additional size-aware criterion biases the algorithm toward more balanced agglomerations.

LOUVAIN. LOUVAIN is a multilevel greedy algorithm that replaces CNM’s community-pair merging with local vertex moves. Instead of selecting pairs of communities (c_i, c_j) to merge based on $\Delta Q_{c_i, c_j}$, Louvain iteratively moves individual vertices u to neighboring

communities c whenever the induced modularity gain $\Delta Q_{u \rightarrow c}$ is positive. Once no improving vertex move exists, each community is contracted into a supernode and the process is repeated on the induced graph. Each local-moving phase runs in linear time in the number of edges, and the number of multilevel passes is typically small, yielding linear overall runtime in practice.

LEIDEN. LEIDEN extends LOUVAIN by adding a refinement phase that splits and restructures communities before contraction to address the possibility of disconnected or weakly connected communities produced by LOUVAIN. After the local-moving phase, each community is refined independently by starting from singleton nodes and merging vertices only with subcommunities inside the original community, provided this yields a positive modularity gain and sufficient internal connectivity. Refinement merges are non-greedy and selected probabilistically among improving options, enabling broader exploration of the search space while never decreasing modularity. As a result, Leiden guarantees that all output communities are connected and locally optimal with respect to vertex moves, while preserving the multilevel modularity-maximization framework.

4.1.2 Non-modularity baselines (random-walk / flow objectives).

WALKTRAP. WALKTRAP is a hierarchical clustering algorithm that departs from modularity optimization and instead operates on the intuition that random walks on a graph tend to get "trapped" into densely connected regions corresponding to communities. Each vertex is characterized by the probability distribution over vertices reached by a t -step random walk starting from it, and structural similarity between vertices (and communities) is defined by comparing these distributions. A distance defined on these distributions yields a computable similarity measure between vertices and communities. Communities are then obtained by agglomerative merging of the closest pairs under this distance, producing a dendrogram similar to CNM. The algorithm runs in $O(mnd)$ time, where d is the height of the resulting dendrogram.

INFOMAP. Instead of comparing random-walk distributions as in WALKTRAP, INFOMAP seeks a clustering that minimizes the expected description length of a random walk, as formalized by the map equation. This objective is rooted in the minimum description length principle, treating community detection as a data compression problem: the optimal partition is the one that allows for the shortest possible "code" to describe the walker's path. INFOMAP's search mechanism is a greedy stochastic multilevel heuristic inspired by the LOUVAIN algorithm, but it incorporates additional fine-tuning and coarse-tuning steps to avoid local minima, similar to the refinements later introduced by LEIDEN. The algorithm operates in two primary phases: a two-level phase that starts with singleton nodes and iteratively moves them to neighboring modules to reduce codelength, followed by a multilevel phase that recursively groups these modules to identify hierarchical structures.

4.1.3 Metaheuristics and ensemble/hybrid search.

CGGC1-RG. CGGC1-RG combines a randomized greedy modularity optimizer with an ensemble-based consolidation scheme known as Core Groups Graph Clustering (CGGC). CGGC extracts *core*

groups of vertices that are repeatedly assigned to the same community across multiple high-quality clusterings and contracts them into supernodes to stabilize and refine the solution. The underlying Randomized Greedy (RG) algorithm follows a CNM-style hierarchical merging process, but restricts merge selection by sampling a fixed number of candidate communities and choosing the best merge among them. Multiple independent RG runs are executed to generate a diverse set of high-quality dendrograms. From these runs, core groups are identified and contracted, and modularity optimization is then repeated on the reduced graph.

VNS. VNS applies a *Variable Neighborhood Search* metaheuristic to modularity maximization. Starting from an initial clustering, the algorithm repeatedly selects a subproblem induced by a randomly chosen community and a growing set of neighboring communities, and perturbs the solution using one of several predefined cluster-level shaking operators (e.g., division, fusion, redistribution). Each perturbed solution is refined using a modularity-based local search (LPAm+), first within the subproblem and, upon improvement, on the full graph. If no improvement is found, the neighborhood size is increased; otherwise it is reset, allowing systematic escape from local optima while maintaining focus on modularity optimization.

COMBO. COMBO is a general-purpose community-detection optimizer that performs a broader search over partitions than LOUVAIN style single-vertex moves. Starting from an initial clustering, the algorithm iteratively considers pairs of source and destination communities and computes the best redistribution of vertices between them. For each such pair, COMBO performs a Kernighan–Lin style sequence of vertex moves, starting from multiple initial configurations (including full merges and partial random transfers), and selects the configuration yielding the largest modularity gain. The best improving recombination is applied at each iteration, and the process repeats until no further improvement in modularity is possible. The algorithm runs in $O(n^2C)$ time, where C is the number of clusters.

VIECLUS. VIECLUS is a memetic (evolutionary) algorithm for modularity maximization that optimizes a population of candidate clusterings rather than a single solution. In a memetic framework, each clustering is treated as an individual, and new solutions are generated by recombination and mutation: recombination exploits common structure in high-quality solutions, while mutation introduces controlled randomness to maintain diversity and escape local optima. VieClus initializes a diverse population using an adapted LOUVAIN method, and generates offspring via *overlay clustering*, where two vertices are grouped together only if they co-occur in both parent solutions. The resulting solution is refined using LOUVAIN or multilevel local search, while mutation is realized by splitting clusters via graph partitioning before recombination. Offspring solutions replace similar population members only if they achieve at least equal modularity, ensuring both solution quality and population diversity. After a set time limit, VIECLUS returns the best clustering found thus far.

4.1.4 Streaming.

HOLLOCOU. HOLLOCOU is a one-pass streaming algorithm for community detection based on the observation that, under random edge orderings, intra-community edges tend to arrive earlier than inter-community edges. The algorithm starts from the singleton clustering and processes edges sequentially. For an arriving edge (u, v) , HOLLOCOU decides whether to merge the communities of u and v based on their current community volumes. If both volumes are below a fixed threshold, the node belonging to the smaller-volume community joins the other; otherwise, no merge is performed. The authors show that this local, volume-based merging rule corresponds to a greedy optimization of modularity in the streaming setting.

CLUSTRE. CLUSTRE follows the node-streaming model unlike the edge-streaming model of HOLLOCOU. Here, nodes arrive sequentially together with their neighborhood lists. Starting from singleton clustering, each node v is assigned to the cluster among its neighbors that maximizes the modularity gain ΔQ , similar to a LOUVAIN local move. During streaming, a quotient graph is constructed whose nodes represent clusters and whose weighted edges summarize inter- and intra-cluster connectivity, preserving modularity exactly. This quotient graph is optionally refined using VIECLUS. The graph may be re-streamed to perform further local improvements, iterating until modularity converges.

4.1.5 Exact Solvers.

BAYAN. BAYAN is an exact algorithm for modularity maximization based on a branch-and-cut formulation of an integer program with binary decision variables encoding community co-membership. It enforces consistent cluster assignments using triangular constraints on node triples, which branch on whether three nodes must belong to the same community or cannot all be clustered together. Upper bounds are obtained by solving linear programming relaxations of the formulation, while lower bounds are computed from feasible clusterings produced by COMBO. Through branching on violated triangular constraints, the algorithm modifies the problem by either contracting node triples into a supernode or discouraging their co-assignment until optimality is certified or a prescribed gap is reached.

MAXSAT. The MAXSAT approach formulates modularity maximization as a weighted partial MaxSAT problem. Binary decision variables indicate whether pairs of vertices belong to the same community, while hard clauses enforce transitivity constraints to ensure a valid clustering. Weighted soft clauses encode pairwise contributions to modularity, such that maximizing the total satisfied weight is equivalent to maximizing modularity. To improve scalability, the formulation uses a sparse encoding that omits redundant transitivity constraints without affecting optimality. Solving the resulting MaxSAT instance yields a clustering with provably optimal modularity.

4.2 Learning-Based Methods

4.2.1 Differentiable modularity maximization. DMO_N (Deep Modularity Networks) introduces an unsupervised modularity-based objective and realizes it using a graph neural network (GNN) for graph clustering. A GNN followed by a softmax function produces

soft cluster assignments, which are optimized using a fully differentiable objective whose main term corresponds to modularity in spectral form. The objective further includes a *collapse regularization* term that prevents trivial solutions in which all nodes are assigned to the same cluster.

DGCLUSTER. DGCLUSTER is a fully differentiable method for deep graph clustering based on graph structure and node attributes. Unlike most learning-based approaches including DMO_N, DGCLUSTER does not require the number of communities to be predefined. It first computes node embeddings using a GNN that incorporates graph structure and node attributes. Pairwise similarities between node embeddings are then interpreted as soft community memberships and used to parameterize the modularity objective. This objective is optimized in a differentiable manner to train the GNN parameters, after which final community assignments are obtained by clustering the learned node embeddings using another algorithm called BIRCH. The overall complexity scales linearly with the graph size.

LIM. LIM (Less Is More) is a learning-based community detection method that integrates global structural information, local graph structure, and node attributes within a modularity maximization framework. It first performs community pre-detection using a LOUVAIN to extract global structural communities, which are summarized as structural community centers in a feature space. Additionally, a graph convolutional network is used to learn node representations that capture local structural information and node attributes. Soft cluster memberships are then computed by measuring similarity between node representations and the structural community centers, integrating global structure with local structural and attribute information. Finally, a modularity maximization objective defined over these soft cluster memberships is optimized to obtain the final community assignment.

4.2.2 Contrastive / infomax objectives aligned with modularity.

COMMDGI. COMMDGI (Community Deep Graph Infomax) is a representation-learning approach to community detection that learns community-aware node representations by maximizing mutual information. Building on Deep Graph Infomax, it jointly maximizes *graph mutual information* between node and graph representations and *community mutual information* between node and community (subgraph) representations to encode the graph and clustering feature of each node. A trainable clustering layer is integrated into the GNN, enabling end-to-end training for community detection. Unlike learning-based methods that directly optimize modularity (e.g., DMO_N, DGCLUSTER, LIM), modularity is used here as an auxiliary objective together with mutual information, via a soft k -means-style clustering layer, to further encourage structurally coherent communities.

UCoDE. UCoDE (Unified Community Detection) is a GNN-based community detection method that jointly supports non-overlapping and overlapping communities within a single unsupervised framework. Like DMO_N and DGCLUSTER, it represents communities through soft cluster assignments produced by a GNN, but unlike these methods it does not rely on direct modularity maximization alone. Instead, UCoDE optimizes a single contrastive objective

derived from the modularity matrix, which simultaneously encourages high intra-community similarity and low inter-community similarity at the community level. This contrastive formulation allows UCoDE to recover non-overlapping communities while naturally supporting overlap, without requiring separate objectives or post-processing. Compared to combined-objective methods such as COMM-DGI, UCoDE uses a single, interpretable loss and requires minimal hyperparameter tuning.

MAGI. MAGI establishes a formal connection between modularity maximization and graph contrastive learning by showing that modularity naturally defines positive and negative node pairs via the sign of the modularity matrix. Specifically, node pairs with positive modularity coefficients are treated as positive samples, while pairs with non-positive coefficients are treated as negatives, yielding a contrastive objective equivalent to relaxed modularity maximization. Based on this insight, MAGI uses modularity maximization as a community-aware pretext task to train a GNN encoder with a softmax-based SimCLR contrastive loss, without relying on graph augmentations. To achieve scalability and capture high-order proximity, MAGI constructs mini-batch modularity matrices using a two-stage random-walk sampling procedure.

5 Evaluation Methodology

In the following section, we present the evaluation methodology used in this experimental survey. The main contribution is the adaptation of 18 clustering algorithms behind one unified interface. In addition to enabling the experiments presented in this survey, it will also serve as an openly available resource for future work on graph clustering, simplifying the evaluation of older methods on new data or machines. This section also includes details on the datasets we used to evaluate the clustering algorithms and the computational environment.

5.1 Implementation

As shown in Table 1, over half of the clustering algorithms in the literature lack ready-to-use interfaces for clustering new data. Furthermore, among those with properly documented code, there are wide differences in the input graph format, output formats, and the additional information they report, such as time and memory consumption. As a result, it can take significant effort to get a particular algorithm to run and extract the results.

To address this problem, we aim to provide an interface with a single common input and output format and consistent time and memory monitoring. The interface exposes a common set of inputs, while allowing solver-specific parameters to be provided when applicable.

$$\text{CLUSTER}(G, T_{\max}, p, k, M_{\max}, \mathcal{F})$$

Where G is the input graph, $T_{\max}$ is the wall-clock time limit, p is the number of threads available to the algorithm, k is the number of clusters to compute, $M_{\max}$ is the memory limit in gigabytes, and $\mathcal{F}$ are the graph features.

Parameters p , k , and $\mathcal{F}$ are only used by a subset of the solvers. From Table 1, parallel solvers use p , those that compute a fixed number of clusters use k , and any solver that uses graph features can optionally accept $\mathcal{F}$. When $\mathcal{F}$ is not provided, but the solver

Table 2: Real-world instances with known ground truth clusters. The $|C|$ columns indicates the number of clusters.

Instance	n	m	$ \mathcal{F} $	$ C $
Amazon-Computers	13 752	245 861	767	10
Amazon-Photo	7 650	119 081	745	8
Citeseer	3 327	4 552	3 703	6
Coauthor-CS	18 333	81 894	6 805	15
Coauthor-Physics	34 493	247 962	8 415	5
Cora	2 708	5 278	1 433	7
ogbn-Arxiv	169 343	1 157 799	128	40
ogbn-Products	2 449 029	61 859 012	100	47
Pubmed	19 717	44 324	500	3
Reddit	232 965	57 307 946	602	41

expects features, a single 1.0 value is assigned to each vertex as the feature.

Regarding code changes, we have tried to make as few modifications to the original source code as possible. Whenever possible, we use the authors' implementations or the most commonly used library implementation. The exact setup for each algorithm is described in our openly available GitHub repository¹. We did, however, make some deliberate changes to ensure consistent timing and quality measures.

- Parsing the input and writing the output to a file is not counted in the timing of the algorithm.
- For search-based algorithms that always run for the entire $T_{\max}$ duration, the recorded computation time is defined as the point in time when the best solution was found.
- Learning-based methods that train a neural network as part of the algorithm will break the training loop at $T_{\max}$, and use the current model at that time to produce the final clustering.

5.2 Dataset

We consider three datasets commonly used in the literature. The first set consists of real-world instances with known ground truth clusters. These graphs are commonly used by the machine learning algorithms. Next, we consider the instances used as part of the 10th DIMACS implementation challenge on graph clustering. This dataset is popular among the classical modularity optimization methods. Finally, we use a dataset of syntetically generated graph with node features. The particular generate we use allow us to explore the algorithms sensitivity to noise in both the graph and features. (TODO, rewrite this section with more details more details).

5.3 Computational Environment

All experiments are conducted on a machine with an AMD EPYC 9754 128-core processor with 755 GB of memory, running Ubuntu 24.04.3 with Linux kernel 6.8.0-87. For algorithms with compiled software, we use GCC version 13.3.0, and for Python implementations we use Python version 3.12.3.

¹<https://github.com/KennethLangedal/Clustering>

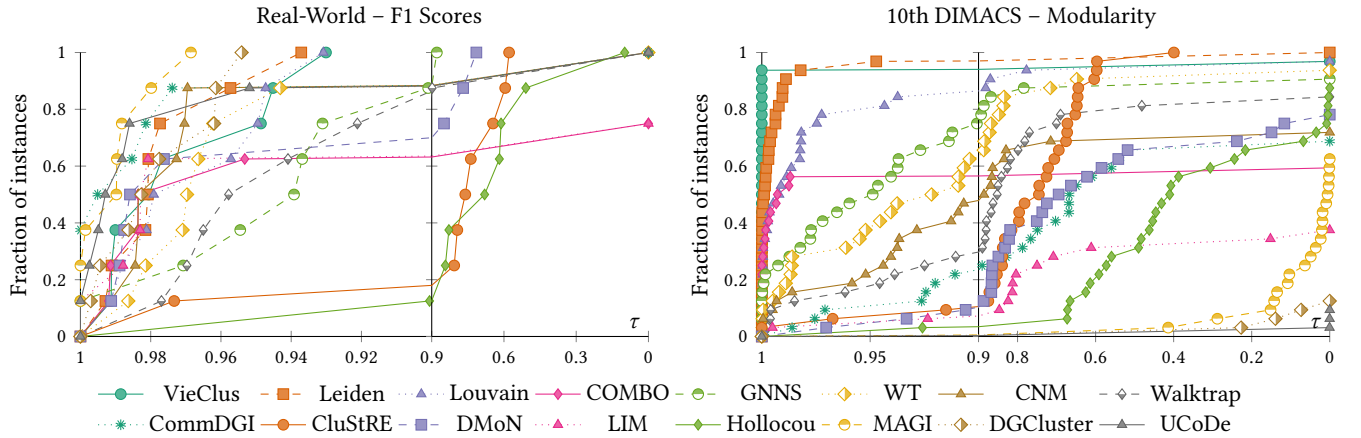


Figure 2: Performance profiles for our first two datasets, the real-world graphs and the instances from the 10th DIMACS implementation challenge. For real-world instances, we use F1 scores to measure solution quality, and for the DIMACS instances, we use modularity. To improve readability, we split the x-axis to highlight the differences between the best algorithms near 1.

In order to speed up experiments, we evaluate three algorithms in parallel. For parallel algorithms, we evaluate them both sequentially and using 16 threads. If an algorithm exceeds 250 GB of memory usage on an input instance, it is killed and recorded as *memory limit exceeded* for that graph. Similarly, we assign each algorithm a time limit of 1 hour per input graph. Unlike the memory limitation, we are more lenient with the time limit due to the nature of some learning based algorithms that train a model for a fixed number of iterations. Only if an algorithm overshoots the time limit by more than 50% do we kill the process and record that instance as *time limit exceeded*.

For algorithms that need to be told the number of clusters, we either provide the exact number when ground truth labels are known, or 16 clusters otherwise, as is commonly used in the literature. Since several of the algorithms are Stochastic, we perform each experiment 5 times and take the best result.

6 Experiments

In this section, we present the results of our experimental evaluation of the 18 algorithms with openly available implementations. This section is divided into three main parts. The first part presents results for the real-world dataset shown in Table 2, and the instances from the 10th DIMACS implementation challenge shown in Table 3. The second part investigates the execution time and memory consumption, as well as the effect that parallel processing has on the algorithms that support it. And finally, the third part presents results for our syntactically generated graphs with varying amounts of noise applied to the graph structure and the node features.

6.1 State-of-the-Art Comparisons

Figure 1 shows the main results of our experimental comparison. The first result that stands out in terms of modularity optimization is that VIECLUS achieves the best result on *all* instances when it finishes within the time and memory constraints. There are two

graphs where VIECLUS exceeded the memory limit: uk-2002 and uk-2007-05, both from the DIMACS set. This result aligns with the results presented by the authors of VIECLUS, who excluded these two graphs in particular because they were too large for a memetic algorithm [6].

(TODO, paragraph on exact solutions).

For graphs with known ground-truth clusters, we observe a much more even split between the algorithms. However, the MAGI method stands out as the best-performing algorithm on this dataset, with consistently high F1 scores and the best worst-case performance. It is worth noting that most algorithms optimize modularity, while the ground-truth clusters could have relatively low modularity. For instance, the ground-truth clusters for the Cite-seer graph have a modularity score of 0.54, much lower than the highest recorded modularity by VIECLUS of 0.89. In fact, all tested algorithms compute clusterings with higher modularity scores than the ground-truth clustering on this graph. One reason for this discrepancy is how the ground-truth clusters are defined. The Cite-seer graph is a citation network in which vertices are scientific publications and edges represent citations. The ground-truth clusters represent categories of each publication. As Figure 1 would indicate, there is no guarantee that a particular publication favors citations within its own category, which would have aligned perfectly with the modularity objective.

There are significant differences between learning-based methods when applied to data without vertex features. Algorithms such as UCoDe, which was among the strongest learning-based algorithms on real-world instances, struggle to produce meaningful clusterings on DIMACS graphs. DMOON and COMMDDGI perform well despite being designed for graphs with node features and a known number of clusters. Unsurprisingly, GNNS performs best in this category, being the only learning-based algorithm designed for graphs without features.

(Streaming paragraph)

(Exact paragraph)

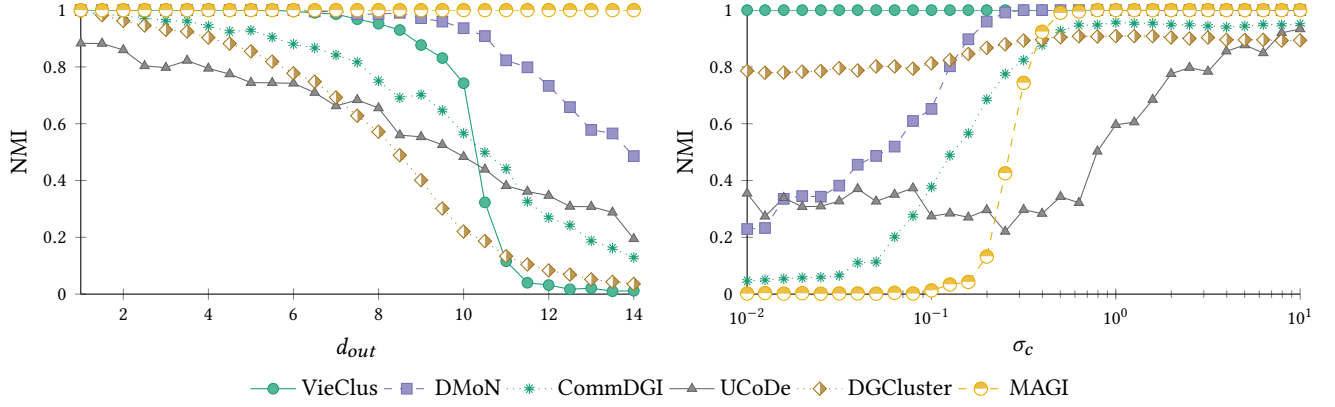


Figure 3: NMI scores on graphs generated with the ADC-SBM graph generator. All graphs have $n = 1\,000$ vertices, $k = 4$ clusters, $d = 20$ average degree, and $\sigma = 1$ intra-cluster center variance. The left plot varies d_{out} while keeping $\sigma_c = 3$, and the right plot varies σ_c while keeping $d_{out} = 2$.

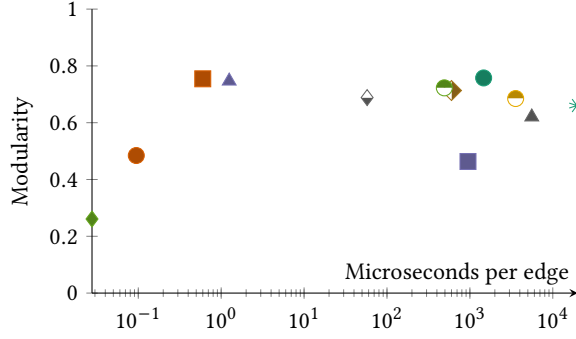


Figure 4: Ratio between time per edge and quality, using the geometric mean for both measures. This plot shares legend with Figure 2. Parallel algorithms have two connected points, showing parallel and sequential versions.

6.2 Computational Efficiency

Judging computational efficiency is made difficult by the wide variety of methods. Figure 4 shows the relation between clustering quality and computational time to highlight the main tendencies of the algorithms. Reading the figure from left to right, the streaming algorithms HOLLOCOU and CLUSTRE are by far the fastest. Next are the two closely related algorithms LEIDEN and LOUVAIN, both scoring near the top in both quality and execution time, only beaten by VIECLUS in terms of solution quality. Unsurprisingly, most learning-based methods take a long time to produce a clustering, but they also achieve high solution quality.

6.3 Syntetic Graphs

To further evaluate the impact and sensitivity of graph features across clustering algorithms, we include experiments with syntactically generated graphs with vertex features. The generator we use is called the attributed, degree-corrected stochastic block model [25]

(ADC-SBM). This generator plants k clusters in the graph, and generates random edges via a distribution that is conditional on those clusters. It does so using the two input parameters d and $d_{out} \leq d$, where d is the expected degree for each vertex and d_{out} is the expected number of neighbors for a node in a different cluster than its own. The ADC-SBM generator also produces node features that are correlated with the community structure. It does so using the two input parameters σ for intra-cluster center variance and σ_c for cluster center variance.

We conduct two sets of experiments with the ADC-SBM generator. First, we generate a sequence of graphs with gradually higher d_{out} , causing the graph structure to carry less information about the original k clusters. And second, we generate graphs with strong community structure as we gradually increase the cluster center variance σ_c , leading the graph features to carry less information about the clusters.

Figure 3 shows the results of our two experiments with the ADC-SBM graph generator. Since this experiment aims to show how feature-aware methods respond to scenarios in which either the graph or the feature signal is distorted, we evaluate only methods that accept graph features as input. For reference, we also include VIECLUS, which performed best among the pure modularity-optimizing methods. When the feature signal is strong, MAGI and DMO-N are the best performers, and at around $d_{out} = 11$, all the feature-aware methods perform better than VIECLUS. It is also clear from this point of view that MAGI relies more on graph features than the graph structure, evident by the strong performance when the graph structure is distorted, and weak performance when the feature signal is distorted.

DMO-N performs well in both scenarios, demonstrating its ability to capture community structure in both the graph and the features without relying too heavily on either one. Similarly, COMM-DGI displays similar performance, but is noticeably worse than DMO-N across the board. DGCLUSTER is somewhat opposite to MAGI, favoring the graph structure over the features. The same holds for UCODE, but to a lesser extent.

7 Conclusion

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Table 3: Graph from the 10th DIMACS implementation challenge. There are no features or ground truth clusters for these graphs.

Instance	<i>n</i>	<i>m</i>
jazz	198	2 742
celegans-metabolic	453	2 025
email	1 133	5 451
PGPgiantcompo	10 680	24 316
adjnoun	112	425
as-22july06	22 963	48 436
astro-ph	16 706	121 251
celegansneural	297	2 148
cond-mat	16 726	47 594
cond-mat-2003	31 163	120 029
cond-mat-2005	40 421	175 691
dolphins	62	159
football	115	613
hep-th	8 361	15 751
karate	34	78
lesmis	77	254
netscience	1 589	2 742
polblogs	1 490	16 715
polbooks	105	441
power	4 941	6 594
cnr-2000	325 557	2 738 969
eu-2005	862 664	16 138 468
in-2004	1 382 908	13 591 473
uk-2002	18 520 486	261 787 258
uk-2007-05	105 896 555	3 301 876 564
road-central	14 081 816	16 933 413
road-usa	23 947 347	28 854 312
chesapeake	39	170
caidaRouterLevel	192 244	609 066
preferentialAttachment	100 000	499 985
smallworld	100 000	499 998
G-n-pin-pout	100 000	501 198

A Instances

Test.

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